Gold Layer Dimensional Modeling

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Uber Project: Gold Layer and Dimensional Data Modeling

1. Introduction to the Gold Layer

The **Gold layer** is the final and most critical stage of the data pipeline. Its primary objective is to deliver data models optimized for business consumption. In this project, we transition from the "One Big Table" (OBT) in the Silver layer to a **Dimensional Data Model** (Star Schema), consisting of **Fact** and **Dimension** tables.

2. Dimensional Modeling Strategy

Dimensional modeling organizes data into two main categories: * **Fact Tables**: Contain quantitative, numerical data (measures) that can be aggregated (e.g., total fare, distance, duration). * **Dimension Tables**: Contain contextual, descriptive data (attributes) that provide meaning to the facts (e.g., passenger details, driver information, location demographics).

Why Separate Facts and Dimensions?

- **Contextual Clarity**: Dimensions provide the "who, what, where, and when" of the business process.
- **Aggregation Efficiency**: Fact tables are optimized for high-speed numerical calculations.
- **Business Standards**: Standardizing dimensions (like IDs and dates) ensures consistency across the entire organization.

3. Implementing Change Data Capture (CDC) and SCD

To maintain accurate historical records, we use **Change Data Capture (CDC)** and **Slowly Changing Dimensions (SCD)**.

3.1. SCD Type 1: Overwrite (Dim Passenger, Driver, Vehicle, Payment)

For attributes where history is not required (e.g., updating a phone number), we use **SCD Type 1**. This method overwrites old values with new ones, ensuring the table always reflects the most current state.

3.2. SCD Type 2: Historical Tracking (Dim Location)

For attributes where tracking changes over time is critical (e.g., city names or regions updating), we use **SCD Type 2**. This method adds a new row for every change, logging the history with timestamps. * **Example**: If "Philadelphia" is updated to "New Philadelphia," SCD Type 2 creates a new record with the same primary key but a unique timestamp, allowing for "as-of" historical reporting.

4. Implementation: Gold Layer Pipeline

The Gold layer is built using **Delta Live Tables (DLT)** views and the `create_auto_cdc_flow` function to automate the ingestion from the Silver OBT into the final dimension tables.

4.1. Example: `dim_passenger` Implementation

```
from pyspark import pipelines as dp


@dp.view

def dim_passenger_view():

    # Read from Silver OBT and select relevant columns

    df = spark.readStream.table("silver_obt")

    df = df.select("passenger_id", "passenger_name", "passenger_email",
```

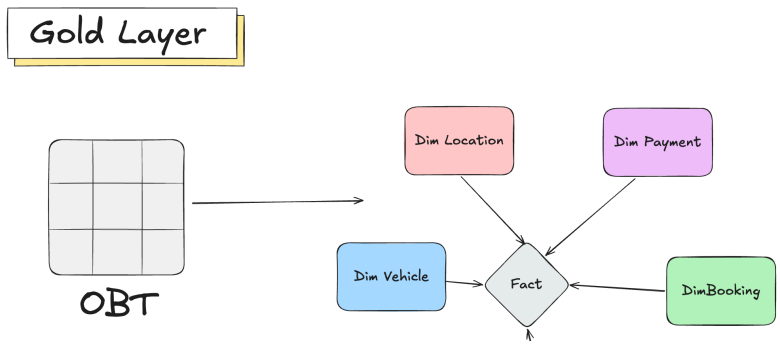
```
    df = df.select("passenger_id", "passenger_name", "passenger_email",
"passenger_phone")

    # Deduplicate based on ID (a passenger may have multiple ride records)

    df = df.dropDuplicates(subset=['passenger_id'])

    return df


# Create the target streaming table and define the CDC flow

dp.create_streaming_table("dim_passenger")

dp.create_auto_cdc_flow(

  target = "dim_passenger",

  source = "dim_passenger_view",

  keys = ["passenger_id"],

  sequence_by = "passenger_id",

  stored_as_scd_type = 1, # SCD Type 1: Overwrite

)
```

4.2. Dimensional Tables Summary

The following tables are created in the Gold layer:

Table Name	Type	Key	Purpose
`dim_passenger`	SCD Type 1	`passenger_id`	Stores unique passenger contact details.
`dim_driver`	SCD Type 1	`driver_id`	Stores driver credentials and ratings.
`dim_vehicle`	SCD Type 1	`vehicle_id`	Stores vehicle make, model, and color.
`dim_payment`	SCD Type 1	`payment_method_id`	Stores payment methods and auth requirements.
`dim_booking`	SCD Type 1	`ride_id`	Stores ride-specific IDs and timestamps.
`dim_location`	**SCD Type 2**	`pickup_city_id`	Tracks historical changes in city/region data.

5. Verification of the End-to-End Flow

To verify the Gold layer, new bookings can be generated via the `api.py` producer. The data flow can be observed as it progresses through the entire Medallion architecture: 1. **Bronze**: Raw events ingested into `rides_raw`. 2. **Silver**: Data parsed, unified, and enriched into `silver_obt`. 3. **Gold**: Enriched data is deduplicated and distributed into the final dimensional tables (`dim_*`), with CDC handling any updates or historical logging automatically.

With the completion of the Gold layer dimensional model, the Uber Data Engineering project is successfully finalized, providing a robust, scalable, and business-ready data warehouse.